Ritesh Mishra

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Education

National Institute of Technology, Hamirpur **2022 – 2026**

Bachelor of Technology in ***Computer Science and Engineering*** *CGPA:* ***9.2/10***

Experience

Research Intern – Robotics and Machine Learning **May – July 2025**

Center for Artificial Intelligence and Robotics (CAIR), IIT Mandi

- **Tech Stack: ROS2**, **PyTorch**, **PyBullet**, **NumPy**, **TensorFlow**
- Developed advanced simulation environment for **2-link robotic manipulator** with **teleoperation control** and automated trajectory recording using **Python** and robotics libraries
- Implemented and trained **deep neural networks** (**MLP**, **LSTM**, **Transformer**) for **multi-step trajectory prediction** over fixed time horizons, achieving superior temporal coherence in motion planning
- Achieved **85+ percent goal-reaching success rate** with **40% reduction in error accumulation** compared to naive one-step prediction methods through horizon-based planning approach
- Conducted comprehensive performance evaluation analyzing trajectory accuracy, smoothness metrics, and model generalization across unseen targets and initial robot states

Projects

Depression Detection using Deep Learning | ***Python, PyTorch, RoBERTa, CNN, WHISPER*** **GitHub**

- Designed comprehensive pipeline processing **1,000+ DAIC-WOZ samples**, detecting **depression indicators** from audio and text cues using advanced **deep learning techniques**
- Achieved up to **80% detection accuracy** by implementing **multi-modal analysis** combining **speech recognition** and **natural language processing**
- Integrated **WHISPER speech recognition** with **RoBERTa-based text analysis** and **CNN for audio feature extraction**, creating robust **multi-modal depression detection system**

Mobile Phone-Based Robotic Arm Teleoperation System | ***ROS2, Robotics, Android*** **GitHub**

- Developed intuitive **mobile interface** using **Android/web app** with **multi-modal input methods** (touchscreen, **IMU sensors**, camera) for **real-time robotic arm teleoperation** and control
- Implemented robust **communication pipeline** between mobile device and robotic system using **Wi-Fi**, **WebSocket**, and **Bluetooth protocols** integrated with **ROS2 framework**
- Built **real-time control system** enabling non-expert users to perform complex manipulation tasks including remote inspection, object manipulation, and assisted navigation

Technical Skills

- **Programming Languages: C/C++**, **Python**, **JavaScript**, **HTML/CSS**, **SQL**, **Assembly (8085)**
- **Frameworks & Libraries: ROS2**, **PyTorch**, **TensorFlow**, **NumPy**, **Pandas**, **Matplotlib**, **SciPy**, **ReactJS**,
- **Specializations: Data Structures & Algorithms**, **Machine Learning**, **Robotics**, **Imitation Learning**
- **Tools & Technologies: Git**, **Linux**, **Database Management**,**Operating Systems**, **Computer Network**,
- **Coursework: Analysis and Design of Algorithms**, **DBMS**, **Computer Networks**, **Compiler Design**, **Artificial Intelligence**,**Theory of Computation**

Achievements

- **LeetCode:** Rated **1762** (**Top 9.25% globally**), Ranked **494 globally** in Weekly Contest 448
- **CodeChef:** Maximum Rating **1441** (**2-star**), Consistent competitive programming participation
- **GATE 2025:** Scored **35.11/100** in Computer Science and Information Technology
- **GATE 2026:** Scored **36.22/100** in Computer Science and Information Technology
- **GATE 2026:** Scored **28.33/100** in Data Science and Artificial Intelligence
- **Research Selection:** Selected for **Research Internship (2025)** at **IIT Mandi** among competitive applicants
- **Research Selection:** Selected for **Direct PhD(2026)** at **IIT BHU** among competitive applicants